A quantitative recoil calibration for finite-gravity comparisons

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Abstract

We bound the finite-principal-series correction to the confluent pure-recoil kernel of Chen, Stanford, Tang and Yang. A positive integral gives a relative window and a nontrivial crossover. We then compare the limiting function with the inverse transform of the reduced semiclassical action used by Chandra, Liska and Post. Their leading actions agree along the pure-recoil route, but their normalized prefactors differ. These are reduced-model calculations, not a finite-gravity correlator. The arguments are author self-reviewed drafts; no priority or independent certification is claimed.

1 Objects and order of limits

Set the de Sitter radius to one and use observer proper time t. CSTY [1], Section 2.6, first takes pure recoil, $Gm_i, G\nu, G\nu' \to 0$, with $m, \nu' \gg 1$, $\nu'/m \ll 1$. Its fixed-average-mass mixed representation and antipodal insertions differ from the finite vacuum packet of our two-dimensional notes. After the small-kink approximation, equation (2.78) defines

$$F_\nu(Q) = {}_3F_2\left(\begin{matrix}1, 1-i\nu, 1+i\nu \\ 3/2, 3/2\end{matrix}; \frac{Q^2}{\nu^2}\right), \quad Q = A\cosh t, \quad A = \frac{\nu\nu'}{2m}. \tag{1}$$

We take $\nu > 0$, $0 \le Q < \nu$, on the real branch below the first branch point. Finite ν here refers to this reduced expression; it does not restore the errors omitted before (2.78). Equation (2.79) takes $\nu \to \infty$ at fixed Q, giving

$$H(Q) = {}_1F_2(1; 3/2, 3/2; Q^2). \tag{2}$$

Only this limit is quantified below. No finite-κ Virasoro $6j$, continuation through $Q = \nu$, or microscopic dictionary is assumed.

2 Positive integral and explicit error

Proposition 1. *For $0 < Q < \nu$, write $r = Q/\nu$. Then*

$$F_\nu(Q) = \frac{1}{2Q}\int_0^{\pi/2} \frac{\sinh(2\nu \arcsin(r\cos\theta))}{\sqrt{1 - r^2\cos^2\theta}}\, \mathrm{d}\theta, \tag{3}$$

$$H(Q) = \frac{1}{2Q}\int_0^{\pi/2} \sinh(2Q\cos\theta)\, \mathrm{d}\theta, \tag{4}$$

$$1 \le \frac{F_\nu(Q)}{H(Q)} \le \frac{\arcsin r}{r\sqrt{1-r^2}}\, e^{2\nu(\arcsin r - r)}. \tag{5}$$

Both functions equal one at $Q = 0$.

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Proof. The beta integral with parameters $1, 1/2$ inserts $(1)_n/(3/2)_n$ into the Gauss series, with weight $\mathrm{d}u/(2\sqrt{1-u})$. Substitute $u = \cos^2\theta$ and use the elementary identity already present in CSTY (2.76):

$$ {}_2F_1(1 - i\nu, 1 + i\nu; 3/2; z^2) = \frac{\sinh(2\nu \arcsin z)}{2\nu z\sqrt{1-z^2}}. $$

Absolute convergence for $r < 1$ justifies termwise integration. The limiting integral follows at fixed Q by dominated convergence, or by its hyperbolic-sine series. These are standard beta-integral operations [3].

For $x \geq 0, c \geq 1$, $\sinh(cx) \leq ce^{(c-1)x}\sinh x$. Indeed, use $\sinh x/x = \int_0^1 \cosh(xu)\,\mathrm{d}u$ and $\cosh(cxu) \leq e^{(c-1)x}\cosh(xu)$. For each θ, put $z = r\cos\theta$, $x = 2\nu z$, $c = \arcsin z/z$. Both $\arcsin z/z$ and $\arcsin z - z$ increase on $[0, r]$. Bound the remaining factor by $(1-r^2)^{-1/2}$. Positive integrands give the upper bound; $\arcsin z \geq z$ gives the lower bound. □

Let $B(Q, \nu)$ be the logarithm of the right-hand side of (5). For $r \to 0$,

$$ B = \frac{Q^3}{3\nu^2} + \frac{2Q^2}{3\nu^2} + O\left(\frac{Q^5 + Q^4}{\nu^4}\right). \tag{6} $$

Thus $0 \leq F_\nu/H - 1 \leq e^B - 1$ is a computable analytic bound for these reduced functions, and

$$ Q = o(\nu^{2/3}) \implies F_\nu(Q)/H(Q) \to 1. \tag{7} $$

This controls relative error of a possibly exponentially large function.

3 Crossover before the branch point

Proposition 2. *Fix $r_0 < 1$. As $Q \to \infty$, uniformly for $0 < r = Q/\nu \leq r_0$,*

$$ F_\nu(Q) = \frac{\sqrt{\pi}}{8Q^{3/2}}(1-r^2)^{-1/4}e^{2\nu \arcsin r}\,[1 + O_{r_0}(Q^{-1})], \tag{8} $$

$$ H(Q) = \frac{\sqrt{\pi}}{8Q^{3/2}}e^{2Q}[1 + O(Q^{-1})]. \tag{9} $$

For any fixed $c > 0$, $Q = c\nu^{2/3}$ consequently gives

$$ F_\nu(Q)/H(Q) \longrightarrow e^{c^3/3}. \tag{10} $$

Proof. The positive exponential in (3) is $e^{Qf_r(\theta)}$, where $f_r = 2\arcsin(r\cos\theta)/r$, continuously extended at $r = 0$. Its unique maximum is at zero, with $f_r''(0) = -2/\sqrt{1-r^2}$. The amplitude and fixed derivatives of f_r are uniformly bounded near zero for $r \leq r_0$. Also $f_r(0) - f_r(\theta) \geq 2(1 - \cos\theta)$, giving a uniform exponential gap outside a fixed neighbourhood. On the scale $\theta = O(Q^{-1/2})$, the even Taylor expansion, Gaussian domination and that tail bound yield relative remainder $O_{r_0}(Q^{-1})$. The negative exponential in the hyperbolic sine is $O(Q^{-1})$ absolutely and exponentially smaller than the positive saddle. The half-Gaussian factor is

$$ \frac{1}{4Q\sqrt{1-r^2}}\,\frac{\sqrt{\pi}(1-r^2)^{1/4}}{2\sqrt{Q}}, $$

giving (8). Its $r = 0$ version gives H. At the crossover $r = c\nu^{-1/3}$, $2\nu(\arcsin r - r) \to c^3/3$, and the amplitude ratio tends to one. □

The resulting growing-time overlap inside the reduced model is

$$1 \ll A\cosh t \ll \nu^{2/3}. \tag{11}$$

The lower inequality permits the saddle approximation; the upper one controls confluence at the function level. At large positive t, the upper condition is $Ae^t/(2\nu^{2/3}) \to 0$, stronger than $Q/\nu \to 0$. The latter can preserve only the leading logarithmic action while losing relative accuracy.

The earlier heavy-background and small-kink limits require separate control for simultaneous scalings of m, ν', ν, t. Even holding the leading kink relation fixed, replacing $\sin(\gamma/2)$ by $\gamma/2$ changes an enhanced exponent on the scale $Q(\nu'/m)^2$. This is a Taylor sensitivity estimate, not a bound on all omitted gravitational effects.

4 A prefactor test of the action-only inverse transform

CLP [2], (3.77)–(3.83) and footnote 12, distinguish their reduced action transform from the full finite-κ amplitude. For $A = \omega_c > 0$, the former is

$$I_A(t) = \int_{\mathbb{R}} \mathrm{d}y \, \exp\left\{2\left[-y\,\mathrm{arsinh}(y/A) + \sqrt{A^2+y^2} - A + yt\right]\right\}. \tag{12}$$

Its unique maximum is $y_* = A\sinh t$. Setting $y = Az$, the exponent is A times a strictly concave function, with second derivative $-2/\cosh t$ at the saddle. Laplace's method, uniformly on bounded real t-intervals, gives

$$I_A(t) = \sqrt{\pi A\cosh t}\, e^{2A(\cosh t - 1)}[1 + O(A^{-1})] \qquad (A \to \infty).$$

The tails decay uniformly on these intervals. Thus

$$\frac{I_A(t)}{I_A(0)} = (\cosh t)^{1/2} e^{2A(\cosh t - 1)}[1 + O(A^{-1})], \tag{13}$$

$$\frac{H(A\cosh t)}{H(A)} = (\cosh t)^{-3/2} e^{2A(\cosh t - 1)}[1 + O(A^{-1})]. \tag{14}$$

Their ratio tends to $(\cosh t)^2$, not one. These $t = 0$ divisions are diagnostic normalizations, not identifications with physical disconnected correlators.

A conditional necessary test follows. Suppose a missing reduced-energy prefactor has smooth leading form $p(y/A)$, is nonzero at the saddle, and preserves the dominant contour. To reproduce the normalized recoil prefactor one would need, on the tested saddle interval,

$$\frac{p(\sinh t)}{p(0)} = (\cosh t)^{-2}, \qquad \frac{p(z)}{p(0)} = \frac{1}{1+z^2}. \tag{15}$$

This is not a derivation or fitted prescription for a Virasoro measure. External legs, spectral weights, determinants and other saddles may change the premise. It gives a concrete test for a later calculation.

CLP has $A = \pi\nu\nu'/(2\kappa\sin\theta_H)$. Along the pure-recoil route $\theta_H = \pi(1 - m/\kappa)$, $A \to \nu\nu'/(2m)$. This degeneration differs from fixed nondegenerate cone angle. No uniform finite-κ error in that degeneration has been established here, so this is not a finite-G match.

5 Numerical checks and next calculation

The script `code/wkb_overlap.py` uses positive integrals with their leading exponent subtracted. Six finite-ν integrals are compared with the defining hypergeometric function and the explicit bound; nine crossover points and six inverse-transform comparisons are also evaluated. All 42 checks pass at 60-digit working precision; crossover and inverse integrals are repeated at 85 and 80 digits. These are empirical arithmetic checks, not interval certification.

For $Q = \nu^{2/3}$, the computed $\log(F_\nu/H)$ is

ν	10^3	10^4	10^6
$\log(F_\nu/H)$	0.337345509	0.334195668	0.333373335

and its predicted limit is $1/3$. At $t = 1, A = 160$, the normalized I/H ratio is 2.380635626; its predicted limit is $\cosh^2(1) = 2.381097846$. No finite list of samples proves the uniform statements.

Next derive the leading prefactor from the continued $6j$, its vertex factors and physical observer insertions, while following the external energy contour. Use (15) as a diagnostic, not an input. The pure-recoil degeneration must also be controlled, or a common finite-deficit regime chosen. The two-dimensional finite-packet theorem does not supply that missing bridge.

References

[1] Y. Chen, D. Stanford, H. Tang and Z. Yang, *Negative shocks versus static patch holography*, arXiv:2607.14042v1, Section 2.6, (2.76)–(2.84).

[2] J. Chandra, D. Liska and B. Post, *3D de Sitter from complex Virasoro TQFT: Observers and knots*, arXiv:2610.30247v1, Section 3.3, footnote 12 and Appendix C.

[3] NIST Digital Library of Mathematical Functions, Section 16.5, beta-integral reductions; Section 15.4, elementary Gauss functions. Online formulas checked 30 October 2026.